RECOVER THE WORKFLOW. PRESERVE THE INTENT.

Every order is a promise.

When an AI workflow stops halfway, reconstruct what happened and verify a recovery that respects the customer.

Gemini reasoning | Controlled execution | Verified outcomes

Workflow Rescue by sustAInX · working local prototype · simulated business services

ONE PERSON. ONE INTERRUPTED ORDER.

The timeout is not the truth.

Maya needs to recover an order for two field kits. The payment request timed out after the charge succeeded. The inventory reservation expired.

> “The exact model matters. One or two extra days is fine. Don't charge us again.”

$498
One existing payment to reconcile

2 kits
The customer's original model

1 change
Stock disappears during recovery

Synthetic scenario. Dollar values illustrate the transaction, not measured business savings.

A BOUNDED REASONING AND EXECUTION LOOP

Gemini proposes. Code verifies.

01 / OBSERVE

Reconstruct the state

Gemini selects payment, stock and dispatch checks. Receipts establish what happened.

02 / ADAPT

Respond to new evidence

A stock-version conflict blocks the initial proposal. Fresh evidence supports a different recovery.

03 / ENFORCE

Respect the mandate

Typed tools, evidence citations, permissions and idempotency constrain every mutation.

04 / VERIFY

Check the ledgers

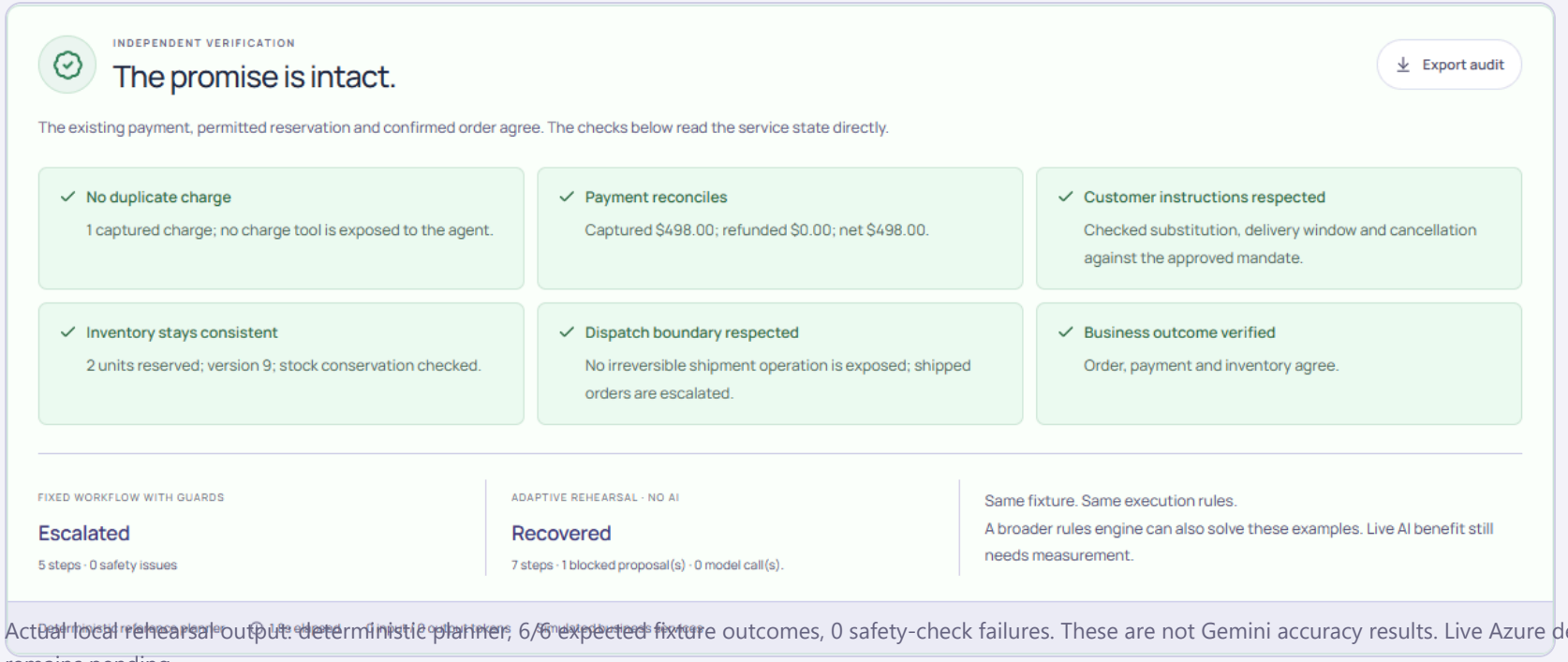
Independent checks verify money, inventory, intent and final order agreement.

No blind second charge. No unapproved substitute. No claimed recovery when the evidence is missing.

WORKING DEMONSTRATION

The promise survives the stock change.

Original items from the next delivery. Existing payment. Evidence for the recovery.



Actual local rehearsal output: deterministic planner, 6/6 expected fixture outcomes, 0 safety-check failures. These are not Gemini accuracy results. Live Azure development separately passed 6/6 fixtures; Gemini validation remains pending.

TECHNICAL DESIGN AND DIFFERENTIATION

Explainable recovery, with boundaries.

The Google stack

- Cloud Run: FastAPI application and web interface.
- Gemini through Vertex AI or the Gemini API: schema-constrained next-step proposals.
- Runtime service identity: Google credentials stay on the server.
- Independent controller: permissions, versions, idempotency and final checks.

What we can defend

- A working observation–replanning loop with execution receipts.
- Explicit escalation when evidence is unknown or dispatch is irreversible.
- Six reviewed scenarios and automated controller/browser checks.
- Inspectable comparison with a guarded fixed workflow.

Retries and compensation are established techniques. A broader rules engine can also solve these fixtures; incremental AI benefit requires live evaluation and wider semantic variation.

FROM PROTOTYPE TO REAL OPERATIONS

Recover the task.
Keep the customer's promise.

Impact to validate

Help enterprise operators resolve interrupted workflows with fewer unnecessary actions and a clearer evidence trail.

Next validation

Live Google runs, broader instruction variants, operator feedback and integration with durable service-owned transaction state.

sustAInX submission handoff

Application: local at localhost:8080
Public Cloud Run URL pending deployment.

Repository: local source ready
Public repository URL pending publication.

Demo: 2:45 recording script ready
Team recording and video URL pending.

All customer data and business services are simulated. No production adoption, measured ROI or contest qualification is claimed.